

Global Facilities - Design & Construction Standard - Mechanical - Specialty Fire Protection System

ECN Facility:	GLOBAL FACILITIES	ECN Area:	ENGINEERING
Owner(s)	ENGINEERING	Target Audience:	All Facilities
Document Type:	Standard	Document ID (pdf):	A3YRXSD74VDV-57553043-463
Version:	2	Last Modified:	05/26/23
Current ECN:	101143870		

[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new Specialty Fire Protection systems as part of new Greenfield Construction or major remodel projects.
- 1.2 The standard is applicable for Specialty Fire Protection systems in new construction or major remodel projects or buildings.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 This document covers new base build and major remodel Specialty Fire Protection System installations only. This document does NOT cover existing Water-Based Fire Protection systems, production tools fire suppression system nor does it require retrofit of existing systems to meet this Standard.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

- **AHJ:** Authority Having Jurisdiction
- **AMHS:** Automated Material Handling System
- **Area per Sprinkler Head:** design term for the floor space that a sprinkler head is designed to protect.
- **Demand Area:** the hydraulically most remote area of the water-based fire protection system. Measured in square feet or square meters.
- **Demand Density:** Water discharge in gpm/min or mm/min
- **FAB:** Semiconductor fabrication facility
- **FATC:** Fire Alarm Terminal Cabinet
- **FM or FMG:** Factory Mutual Global Insurance Company
- **FOUP:** Front Opening Unified/Universal Pod
- **FCT:** Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's
- **GFTT:** Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
- **GFC:** Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- **Global Facilities Central Team:** This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team:** This is used interchangeably with Global Construction Team
- **Global Facilities Operations Support Team:** This is used interchangeably with Global Facilities Operations Team
- **Global Facilities Planning Team:** This is used interchangeably with Global Facilities Line Planning Team
- **HVM:** High Volume Manufacturing facility
- **HPM:** Hazardous Production Materials
- **IBC:** International Building Code
- **ICC:** International Code Council
- **IFC:** International Fire Code
- **k-factor:** a performance metric for the flow of water from a sprinkler head. It is equivalent to the C_v factor for hydraulic pressure drop calculations.
- **LOTO:** Lock-Out, Tag-Out
- **NFPA:** National Fire Protection Association
- **PAFS:** Pre-Action Fire Sprinkler system
- **QR:** Quick Response sprinkler head
- **SR:** Standard Response sprinkler head
- **UL:** Underwriters' Laboratory

4.0 Roles and Responsibilities

Roles	Responsibilities
Global Facilities Engineering Team (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
Regional Facilities Engineering Team (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

The Specialty Fire Protection/Suppression systems covered within this document use an inert gas/ chemical agent or a combination of water and additives to control the spread of a fire within specialty room such as Electrical Room, HPM Room, Computer Room, etc. Refer to **Table 1** for an overview of the applications, locations, and systems covered in this Standard. Guidance is also provided for water-based system types that are covered in separate Standards. In some locations and/or occupancies, more than one system type may be an acceptable option. For example, an Electrical Room shall be protected by a Clean Agent Fire Extinguishing system, when clean agent type is not acceptable by local code or local authorities, then pre-action system to be considered and subject to review and approval. Specialty Room Fire Protection systems will generally provide the best form of fire protection on the most cost-effective basis.

- 5.1 **Specialty – Foam Sprinkler Systems** is a type of wet sprinkler system that combines both water and a foaming agent for flammable/combustible liquids or large-scale fire extinguishment. The system piping is typically filled with water. The foaming agent is added or injected into the water through the discharge device into the water and form foam solution. When the water mixes with the foaming agent, expansion occurs and creates an immense coverage of foam to fill expansive areas. Activation of the system is generally through burst of sprinkler head. This type of system is typically reserved for locations there is a lot of flammable or combustible liquids, such as, Bulk Solvent Room, Solvent Waste Collection Room, etc. The reason that this type of system is most commonly used in these applications is because of the widespread capabilities of foam to not only extinguish but also smother flames in a way that prevents reignition.
- 5.2 **Specialty – Clean Agent Fire Extinguishing System** is a type of fire extinguishing system that use clean agent which is electrically non-conductive, volatile, or gaseous, and that does not leave any residue upon evaporation. The system piping is typically filled with pressurized chemical (FM200, Halon 1301, Novec 1230, etc.) or gas (CO2, Inergen, etc.) with respective reservoir tank. Activation of the fire suppression system riser is generally through two fire detection devices (e.g., heat or smoke detector). This type of system is generally used to minimize property damage and business interruption in processes or occupancies of high value or particular susceptibility to fire, such, IS machine rooms, electrical rooms, etc.

Table 1 – Fire Suppression System Types

Fire Suppression System Type	Typical Spaces Covered by System Type
Specialty Type Systems	
Specialty – Foam	Solvent Rooms (i.e., Organic & Solvent storage or waste, IPA, PGMEA)
Specialty – Clean Agent Fire Extinguishing	- Chemical Dispense Units (CDU's) - On-board production tools - Electrical Rooms (Fully enclosed) - Server Rooms, IS machine rooms - IG-541 and IG-100 are the recommended Clean Agent Extinguishing/Suppression System. IG-541 and IG-100 to be primary selection. - Other extinguishing/suppression agents such as CO2, Halon, Inergen shall NOT be used - Refer to Table 2 for detail recommendation.
Wet Type Systems	
Wet Pipe Systems (Separate BKM)	- FAB Cleanroom, Clean Subfab, Utility Subfab - FAB Plenum or Interstitial levels where required. - Chemical Rooms (Acids, Caustics, etc.) - Gas Rooms - Utility Areas (Central Plant Mechanical Rooms, Water/Wastewater Plant) - Standard Building Spaces (i.e., Offices, Corridors, Lobbies, Toilet Rooms, etc.) - Fire protection inside combustible (i.e., poly vinyl chloride (PVC), polypropylene (PP), fiberglass reinforced plastic (FRP) fume exhaust ductwork
Dry Type Systems	
Dry System (Separate BKM)	- Cooling Towers - Areas subject to freezing conditions - Docks - Unheated warehouses - Epoxy storage coolers
Pre-Action Type (Separate BKM)	Possible alternative for electrical rooms (i.e. Substation, Transformer, Electrical), computer rooms/server rooms, subject to Review and Approval by Global Facilities

6.0 System Design Methods

The piping system must be designed to ensure sufficient pressure exists at any point within the system for adequate flow and pressure to meet stated design requirements and suppress a fire. Piping configuration/arrangement, pipe lengths, and pipe diameter/sizing are the most influential hydraulic design factors.

6.1 Specialty –Foam Sprinkler System

- 6.1.1 Tree or dead-end piping arrangement shall be the default configuration for foam sprinkler systems.
- 6.1.2 Piping carrying foam solution shall be sized as if carrying plain water.
- 6.1.3 Pressure drop (as a function of flow rate) through the system is the primary design consideration for pipe sizing.
- 6.1.4 For the same flow rate, larger pipes exhibit lower pressure drop than smaller pipes. Installation of pipes larger than necessary consumes space and is not cost efficient.
- 6.1.5 See Section 8.6 for details on pressure drop and hydraulic calculation requirements.

6.2 Specialty – Clean Agent Fire Extinguishing System

- 6.2.1 Micron approved Clean Agent Fire Extinguishing System are **IG-541 or IG-100**, which is to be used as primary selection. **Pre-action system may be considered if IG-541 or IG-100 not acceptable** by AHJ and its use is subject to review and approval by Global Facilities.
- 6.2.2 Tree or dead-end piping arrangement shall be the default configuration for clean agent fire extinguishing systems.
- 6.2.3 Piping carrying clean agent shall be sized as if carrying plain water.
- 6.2.4 Pressure drop (as a function of flow rate) through the system is the primary design consideration for pipe sizing.
- 6.2.5 For the same flow rate, larger pipes exhibit lower pressure drop than smaller pipes. Installation of pipes larger than necessary consumes space and is not cost efficient.
- 6.2.6 See Section 8.6 for details on pressure drop and hydraulic calculation requirements.
- 6.2.7 CO2 fire suppression system shall **NOT** be used due to health and safety risk. For any special consideration, formal review and approval is required by GFTT, FCT, Global EHS, Site EHS and Executive Management.

- 6.2.8 In the absence of code direction, the fire sprinkler system shall be designed and installed per the recommended fire protection system for electrical room, data centre and IT rooms as outlined below in **Table 2**.

Table 2 – Recommended Fire Protection System for Electrical Room, Data Centre and IT Rooms

Rooms	Type of Fire Protection	Remarks
Server Room / Data Centre	IG-541/IG-100	
Electrical Room (Fully Enclosed)	IG-541/IG-100	
Electrical Room (Semi-enclosed or Open)	Pre-Action system with HSSD detection	Subject to local fire code and AHJ.
FMCS Room (with SCADA server)	IG-541/IG-100	
FMCS Room (without SCADA server)	Wet Sprinkler	
IDF Room	Wet Sprinkler	Subject to local fire code and AHJ.
MDF Room	Wet Sprinkler	Subject to local fire code and AHJ.
TDR Room	Wet Sprinkler	Subject to local fire code and AHJ.

7.0 System Components

All components within a Specialty Fire Sprinkler system must be approved for their intended use within the system. It is strongly recommended that all components are FM approved or equivalent. In some jurisdictions, such as Japan, FM approved devices are not readily available on a cost-effective basis. GFTT and/or Risk Management should be consulted to develop equivalent options to FM approved devices. The AHJ is the final approving entity for materials and components within a system.

Note: The availability of materials or manufacturers listed will vary from site to site. Local Construction Specifications shall take precedence over any manufacturers listed. The manufacturers listed below are only examples of possible suppliers. Manufacturers beyond those listed may be used at the discretion of the AHJ and the Design Engineer.

7.1 Specialty –Foam Sprinkler System

The terms and components listed below are specific to Foam Fire Sprinkler Systems.

7.1.1 Valves

7.1.1.1 Position Indicating Butterfly Valve (IBV) can be used to isolate an entire zone at a riser room or isolation for a sub-system as an alternate to an OS&Y valve.

- i. Approved Manufacturers - Anvil International, Tyco Fire & Building Products, Victaulic, Viking, or approved equivalent

7.1.1.2 Outside Stem & Yoke (OS&Y) is one type of valve used for isolation of a building or zone on a Fire Protection system. Typically, OS&Y valves are used on a sprinkler system riser.

- i. Approved Manufacturers – Milwaukee Valve Co., NIBCO Inc., Victaulic, Viking, or approved equivalent

7.1.1.3 Ball valves are typically used on piping sizes 2" (50mm) and smaller. A ball valve can be used for isolation of a smaller sub-systems or for ancillary functions on a zone. Ancillary functions may include drains, Inspector's test ports, etc.

- i. Approved Manufacturers - Anvil International, Kennedy Valve, Tyco Fire & Building Products, Viking or approved equivalent

7.1.1.4 Check valves allow flow in one direction only. They are typically installed downstream of a zone isolation valve in the Riser room. Check valves prevent flow across sprinkler zones/systems and protect the quality of the main water supply.

- i. Approved Manufacturers – Reliable Automatic Sprinkler Co, Inc., Tyco Fire & Building Products, Victaulic, Viking, or approved equivalent

7.1.1.5 Monitored valves / tamper switches provide an electronic signal (usually routed back to a Fire Alarm or Life Safety Systems Control Panel) that indicates the status of the valve (i.e., open or closed). Monitoring hardware can be installed on a wide variety of valve listed above, with the exception of a curb box valve.

7.1.2 Foam Concentrate Proportioners

7.1.2.1 Use balanced-pressure or positive-pressure injection methods to introduce foam concentrates into the water flowing through the supply piping to the system. Based on the hazard or occupancy to be protected, use one of the following balanced- or positive-pressure injection methods (reference diagram from FMDS-0412):

- Bladder tank proportioning (**Figure 1**)
- Balanced pressure proportioning (**Figure 2**)
- In-lined balanced proportioning (ILBP) (**Figure 3 and 4**)
- Line proportioning, deluge application only (**Figure 5**)
- Wide Range proportioning (**Figure 6**)
- Variable Viscosity Proportioner (**Figure 7a and 7b**)

Figure 1: Bladder tank proportioning.

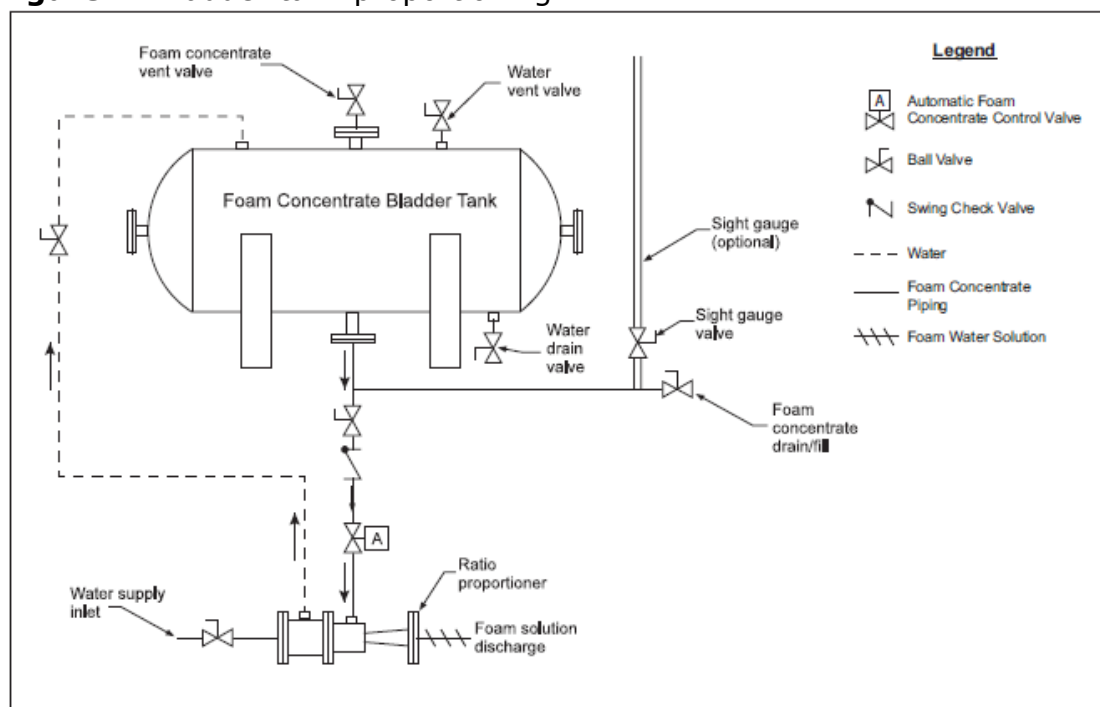


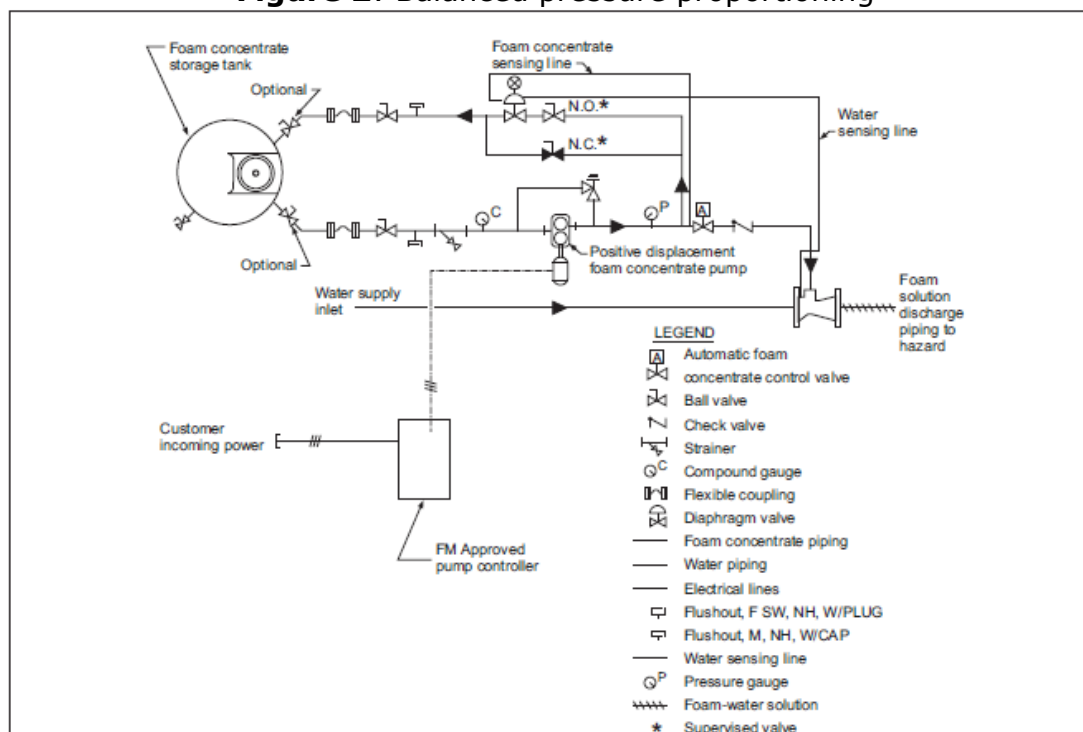
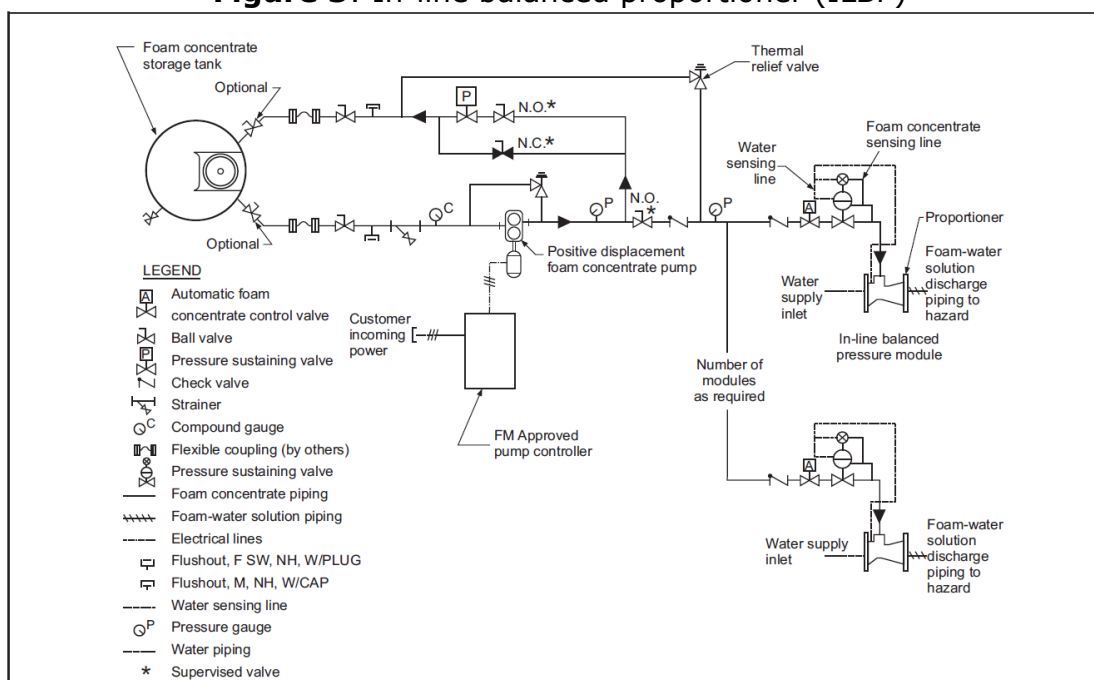
Figure 2: Balanced pressure proportioning**Figure 3: In-line balanced proportioner (ILBP)**

Figure 4: In-line balanced proportioner (ILBP), deluge application

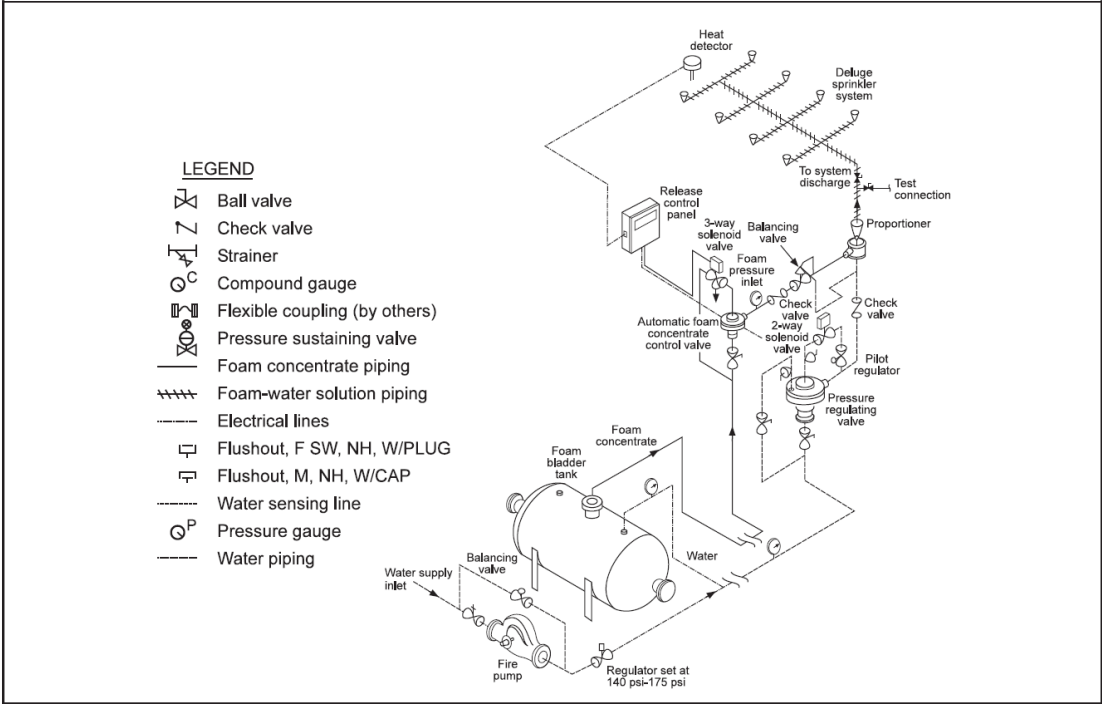


Figure 5: Wide range proportioner

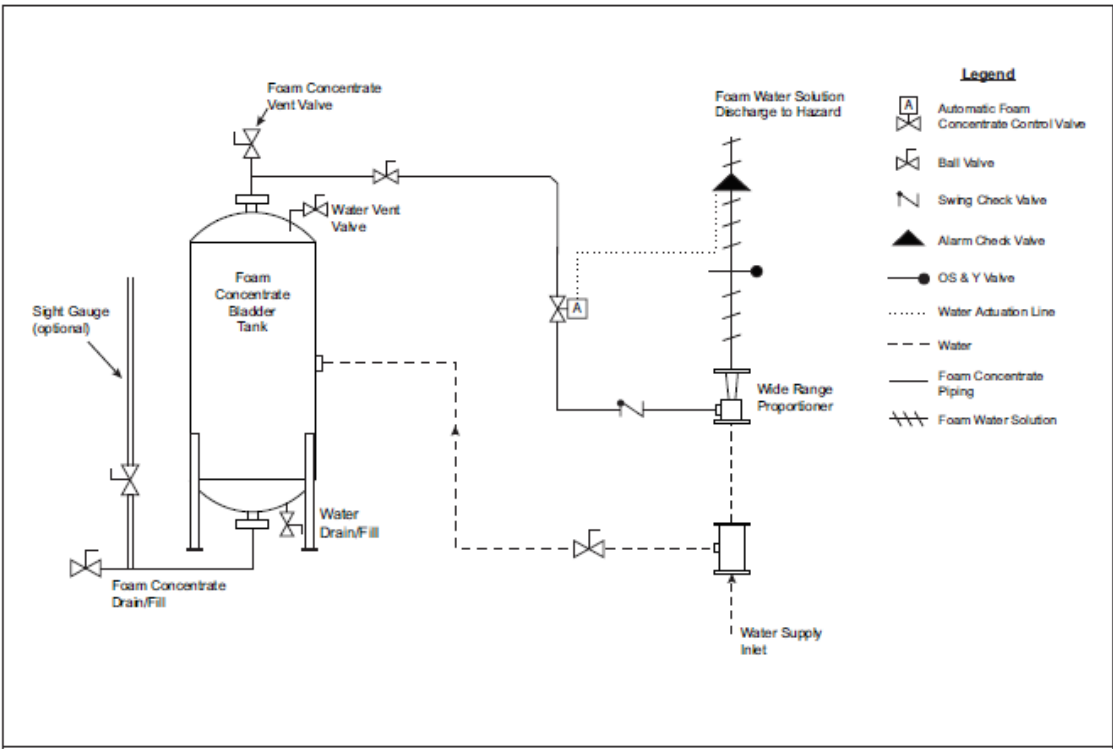


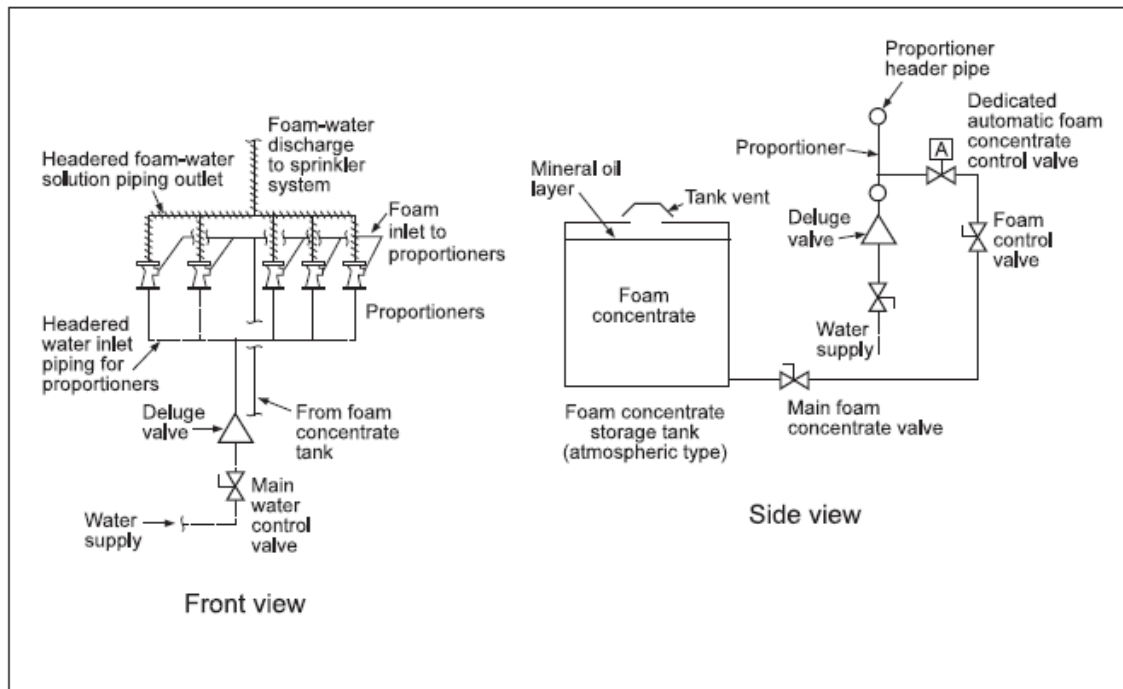
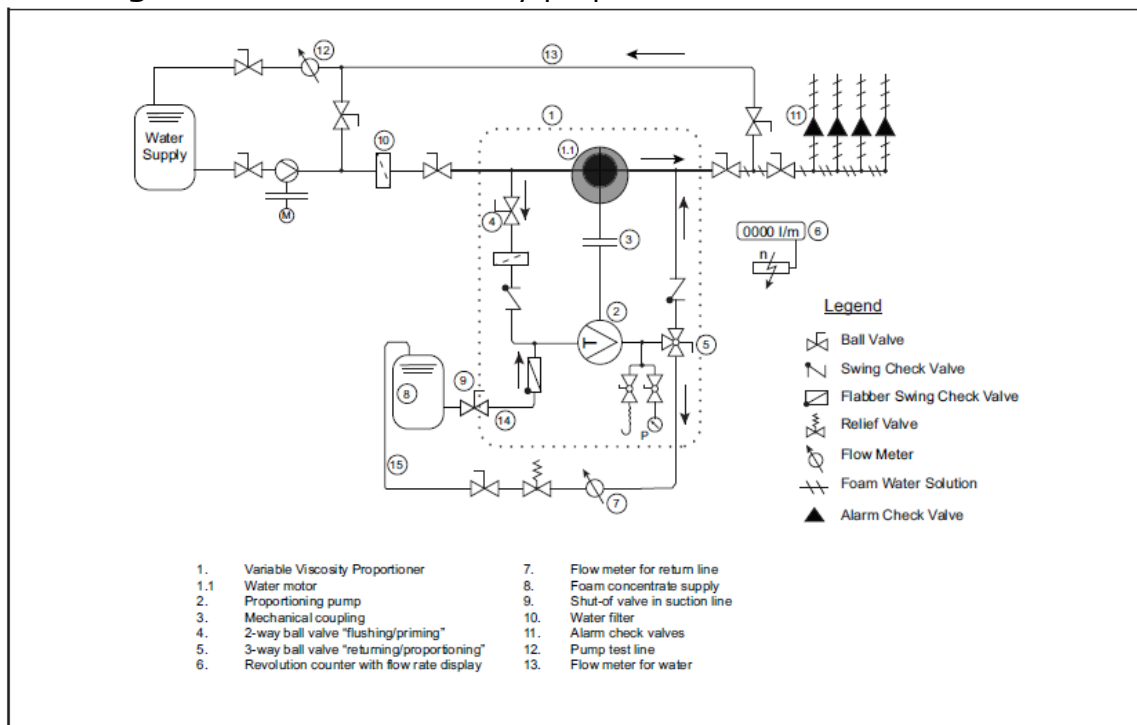
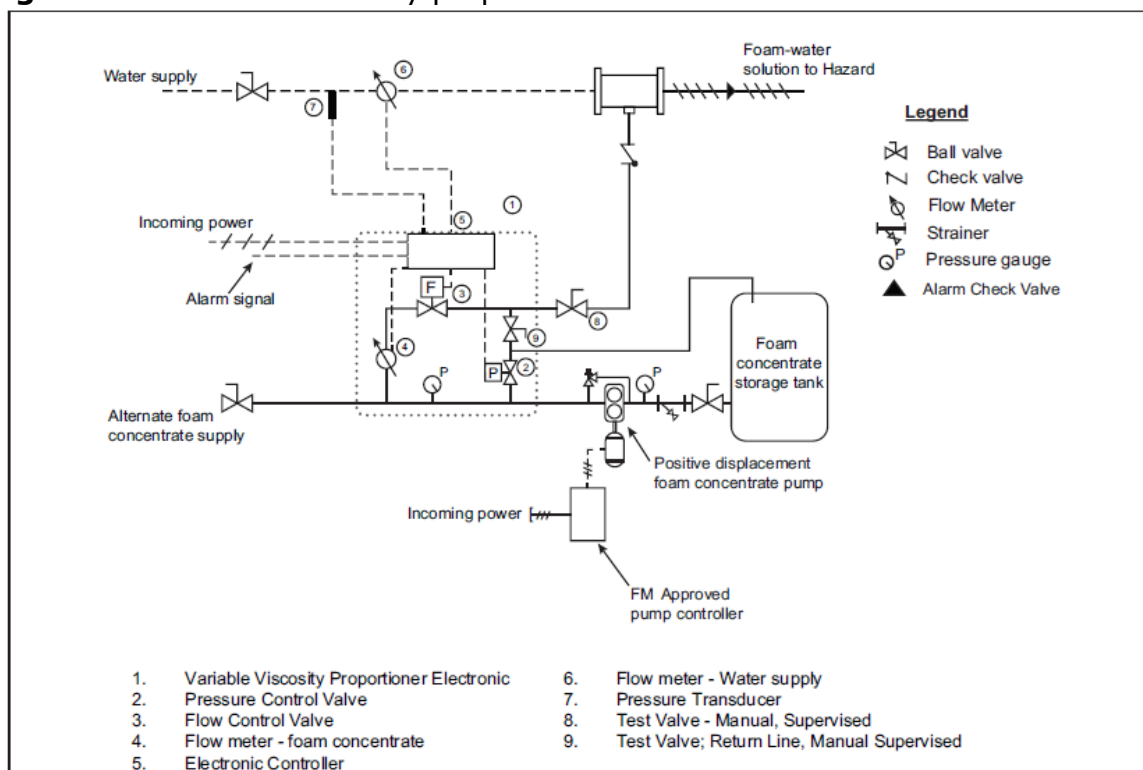
Figure 6: Line proportioning, deluge application only**Figure 7a:** Variable viscosity proportioner: water-driven motor

Figure 7b: Variable viscosity proportioner: electronic controller with flowmeters

7.1.3 Piping

7.1.3.1 All foam concentrate piping shall conform to one of the following standards:

- ASTM A312 (SS 304L & SS 316L)
- ASTM B43 (Red Brass)
- ASTM B315 (Copper Alloy)
- Others approved equivalent by the AHJ

7.1.3.2 Pipe carrying foam concentrate shall not be galvanized.

7.1.3.3 Carbon steel pipe shall be permitted to be used in accordance with the manufacture's certification of compatibility with the foam concentrate and as approved by the AHJ.

7.1.3.4 Piping installed in non-Cleanroom areas shall be black steel.

7.1.3.5 Refer to the project specific Construction Specifications for additional requirements on materials, joining methods, and approved manufacturers.

7.1.4 Fittings

- 7.1.4.1 For piping 2" and smaller, threaded fittings shall be used.
- 7.1.4.2 For piping 2-1/2" and larger, connections may be made via any of the following methods:
 - a) Approved rolled & grooved fittings
 - b) Flanges/bolts
 - c) Welding
- 7.1.4.3 When grooved couplings and fittings with elastomeric seals are used, ensure compatibility with the foam concentrate, foam-water solution, or water, as applicable.
- 7.1.4.4 Refer to the project specific Construction Specifications for additional requirements on materials, joining methods, and approved manufacturers.

7.1.5 Sprinkler Heads

Proper selection of sprinkler heads is of critical importance in any Specialty Room Fire Protection system. In all cases, the sprinkler heads selected must be approved for the application and occupancy.

- 7.1.5.1 k-factor is a numerical representation of the ability of a sprinkler head to deliver water. A higher k-factor indicates higher water flow at a given pressure. Certain occupancies require specific k-factor heads to mitigate an event.
- 7.1.5.2 Temperature Rating indicates the estimated temperature required to fuse/open a sprinkler head.
- 7.1.5.3 Response Type is either Quick Response (QR) or Standard Response (SR) for non-storage applications. A QR sprinkler head is the default for wet pipe systems. Early Suppression Fast Response (ESFR) sprinkler heads may work for certain storage locations.
- 7.1.5.4 The sprinkler head type used must be approved for foam fire extinguishing application and listed for use with specific foam concentrate.
- 7.1.5.5 Protective guard covers shall be provided for all sprinkler heads located above critical areas including, but not limited to Utility Room areas, HPM Areas.
- 7.1.5.6 Re-use of sprinkler heads is forbidden. Only new sprinkler heads may be installed in a system. Once

removed from a system, a sprinkler head shall not be re-installed in any Micron sprinkler system.

- 7.1.5.7 Approved Manufacturers: Viking, Reliable, Tyco, or approved equivalent

7.1.6 High Point Vents

- 7.1.6.1 Oxygen is a primary contributor to corrosion within wet pipe systems. High point vents within a system or zone allows trapped air and non-condensable gases at the air-water interface to escape from a wet pipe system.

- 7.1.6.2 A typical high point vent will utilize two float assemblies plumbed in series with a leak detection device between the two floats. Contact GFTT and/or Risk Management for details associated with the high point vents.

- 7.1.6.3 A high point vent shall be installed at the highest elevation within each zone or sub-system.

- 7.1.6.4 Approved Manufacturers – Engineered Corrosion Solutions (ECS), Potter Electrical Signal, or approved equivalent.



Figure 8: Example of a high point vent assembly

7.1.7 Foam Concentrate

7.1.7.1 The AHJ is the final approving entity for types of concentrate to be used based on local fire or environmental regulation.

7.1.7.2 When protecting hydrocarbon fuels (diesel or hydrocarbon solvent), use one of the following types of foam concentrates in accordance with FM approval listing:

- Aqueous film-forming foam-C6 (AFFF-C6)
- Alcohol-resistant aqueous film-forming foam-C6 (AR-AFFF-C6)
- Synthetic fluorine-free foam (SFFF)
- Alcohol-resistant synthetic fluorine-free foam (AR-SFFF)

7.1.7.3 When protecting polar solvents (solvents with appreciable water solubility or water miscibility, such as methyl alcohol, ethyl alcohol, ethanol and acetone), use an alcohol-resistant (AR) foam concentrate.

Aqueous film-forming foams (AFFF) and synthetic fluorine-free foams (SFFF) will not form films at the interface with polar solvent.

7.2 Specialty – Clean Agent Fire Extinguishing System

The terms and components listed below are specific to Clean Agent Fire Extinguishing Systems.

7.2.1 Storage Container

7.2.1.1 Install new storage container. Do not use repurposed containers.

7.2.1.2 Provide a nameplate on each container that includes the following information:

- Manufacturer and system information
- Clean agent type
- System working pressure
- Allowable ambient storage temperature range
- Agent quantity (tare and gross volume for inert gases)

7.2.1.3 Provide a means to indicate that each container is correctly charged (e.g., pressure gauge or transducer, liquid level indicator, weighing device)

7.2.2 Valves

- 7.2.2.1 All valves shall be listed or approved for intended use.
- 7.2.2.2 All gaskets, O-rings, sealants, and other valve components shall be constructed of materials that are compatible with the agent. Valves shall be protected against mechanical, chemical, or other damage.
- 7.2.2.3 Where directional valves are used for multi-hazard protection, the directional valves shall be listed or approved for use with the installed suppression system.
- 7.2.2.4 Where directional valves are used for multi-hazard protection, the control equipment shall be specifically listed for the number, type, and operation of those valves.

7.2.3 Piping

- 7.2.3.1 All piping shall have physical and chemical characteristics such that its integrity under stress can be predicted with reliability.
- 7.2.3.2 Thickness of the piping shall be calculated in accordance with ASME B31.1. The internal pressure used for this calculation shall not be less than the greater of the following values:
 - The normal charging pressure in the agent container at 21°C (70°F)
 - Eighty percent of the maximum pressure in the agent container at a maximum storage temperature of not less than 55°C (130°F), using the equipment manufacturer's maximum allowable fill density, if applicable.
- 7.2.3.3 For inert gas system incorporating a pressure reducing valve, use a pressure for piping and fitting wall thickness calculation upstream of the valve as indicated in Section 7.2.1.3. Downstream of the pressure reducing valve, use the maximum expected pressure based on the pressure reducing valve's settings.
- 7.2.3.4 Provide automatically operated pressure relief devices where valve arrangements result in closed piping sections. Design the pressure relief devices to operate at a pressure in accordance with the manufacturer's design, installation, operation, and maintenance manual and any applicable national or international codes and standards.

7.2.3.5 Cast Iron pipe, steel pipe conforming to ASTM A210, or nonmetallic pipe shall **NOT** be used. .

7.2.3.6 Refer to the project specific Construction Specifications for additional requirements on materials, joining methods, and approved manufacturers.

7.2.4 Fittings

7.2.4.1 For piping 2" and smaller, threaded fittings shall be used. The use of other fittings such as compression fittings are subject to system manufacturer's specification and approved type of fittings in accordance to AHJ.

7.2.4.2 For piping 2-1/2" and larger, connections may be made via any of the following methods:

- a) Approved rolled & grooved fittings
- b) Flanges/bolts
- c) Welding

7.2.4.3 Fittings shall have a minimum rated working pressure equal to or greater than the minimum design working pressure specified for the clean agent being used.

7.2.4.4 All threads used in joints and fittings shall conform to ASME B1.20.1. Joint compound, tape, or thread lubricant shall be applied only to the male threads of the joint.

7.2.4.5 When grooved couplings and fittings are used to join pipe, the manufacture's pressure and temperature ratings of the fitting shall not be exceeded.

7.2.4.6 Cast-iron fittings shall **NOT** be used.

7.2.4.7 Refer to the project specific Construction Specifications for additional requirements on materials, joining methods, and approved manufacturers.

7.2.5 Discharge Nozzle

Proper selection of discharge nozzle is of critical importance in Clean Agent Fire Extinguishing System. In all cases, the discharge nozzle selected must be approved and listed for intended use.

7.2.5.1 Discharge nozzle, discharge orifices and discharge orifices plates and inserts shall be of a material that is corrosion resistant to the clean agent used and the atmosphere in the intended application.

- 7.2.5.2 Where clogging by external foreign materials is likely, discharge nozzles shall be provided with frangible discs, blowoff caps, or other suitable devices. These devices shall provide an unobstructed opening upon system operation and shall be located so they will not injure personnel.
- 7.2.5.3 Nozzles shall be installed so as to be free of obstructions that could interfere with the proper distribution of the discharged agent in accordance with the manufacturer's installation and maintenance manual.
- 7.2.5.4 Approved Manufacturers: FIKE, Viking, SEVO, Tyco, Sri or approved equivalent

8.0 Design Criteria

8.1 Specialty –Foam Sprinkler System

8.1.1 Occupancy Types and Demand Density

The performance requirements of any foam sprinkler system inside a building are dependent upon the occupancy type, which identifies the nature of the hazard within the space. An occupancy with a high hazard will require a higher water density than an occupancy of a lower hazard.

- 8.1.1.1 The design discharge density shall be in accordance with the applicable occupancy standard for water or foam-water system but in no case less than 0.16 gpm/ft² (6.5 mm/m²).

8.1.2 Sprinklers - Area of Coverage and Head Spacing

- 8.1.2.1 Sprinkler heads are typically aligned to 8'x12' (2.44m by 3.66m) or a 10'x12' (3.05m by 3.66m) spacing pattern.
- 8.1.2.2 See **Table 3** for standard/typical design requirements for nominal Design Demand Area and nominal Area per Sprinkler Head guidelines, as a function of Occupancy.
- 8.1.2.3 Provide the sprinkler heads at proper locations, spacing, and elevations as required to comply with all NFPA, local code, and AHJ requirements for proper sprinkler coverage.
- 8.1.2.4 Coordinate sprinkler head locations with all potential obstructions below, including but not limited to building structure, utilities, equipment, etc., which may affect proper sprinkler coverage.

8.1.2.5 Sprinkler heads shall be located a minimum distance and elevation away from building structure and other obstructions per NFPA (NFPA 16, Section 7.6), local code, and AHJ requirements, to provide proper sprinkler coverage and prevent potential damage to the sprinkler heads.

8.1.3 Performance Specs

8.1.3.1 Applicable more restrictive local code requirements at each location supersede any requirements contained within this document. The performance requirements of any system will be dependent upon the type of occupancy.

8.1.3.2 The system shall be design to deliver foam water solution from the four most remote discharge devices within 2 minutes of system operation.

8.1.3.3 In the absence of code direction, the fire sprinkler system shall be designed and installed per the Standard/Typical Design Requirements as outlined below in **Table 2**.

Table 3: Standard/Typical Design Requirements for Foam Sprinkler Systems

Occupancy	Demand Density, gpm/sqft (mm/m2)	Design Demand Area ² , sqft (m2)	Area per Sprinkler Head ¹ , sqft (m2)
Solvent Room (Subfab)	≥0.16 (≥ 6.5)	5,000 (465)	≤100 (≤ 9.3)
Utility Areas (Diesel Tank Farm, Diesel Storage Area, etc.)	≥0.16 (≥ 6.5)	5,000 (465)	≤100 (≤ 9.3)

¹ Refer to **Section 8.1.5** for specific guidelines regarding hydraulic calculations and area per sprinkler head.

² Where applicable occupancy standards specify design demand areas different from specified in this, the occupancy standards shall take precedence.

8.1.4 Water Supply

8.1.4.1 The source of water for the foam sprinkler system may be from a municipality or an industrial water supply into Site Fire Water Tank.

8.1.4.2 Water supply shall be designed at the design discharge rate and pressure for at least 60 minutes, unless otherwise required by the occupancy standards, AHJ or Local Code.

8.1.5 Demand Area Hydraulic Calculations

Depending on the piping arrangement of the system, the Demand Area can be identified using different methods.

8.1.5.1 All hydraulic calculations (flow vs. pressure loss) shall be performed by an approved hydraulic loss calculation program.

8.1.5.2 Ensure that any hose flow requirement is captured in the hydraulic calculations. A minimum hose flow demand of 250 gpm (950 lpm) is typical for semiconductor fabrication facilities; some more demanding applications, such as, solvent rooms and cooling towers, require up to 500 gpm (1900 lpm) hose flow allowance. Refer to your specific local code or AHJ for each location and application. Contact GFTT and/or Corporate Risk Management for further details.

8.1.5.3 For hydraulic calculations, the designer shall use the maximum area per sprinkler head that is allowed in Code.

8.1.5.4 The maximum area per sprinkler head allowed by Code is usually greater than the nominal design area per sprinkler head. Hydraulic calculations will not use the design area per sprinkler head, nor the actual area per sprinkler head.

EXAMPLE:

In the United States, the IFC limits a sprinkler head within a clean room to a maximum of 130 ft² of coverage area. The ceiling is arranged in a 2' x 4' grid array. The designer chooses to set a nominal design area per sprinkler head in an 8' x 12' array to align with the ceiling grid. The nominal design area is now 8' x 12' = 96 ft² per sprinkler head. The hydraulic calculations will be run using the 130 ft² per sprinkler head.

8.1.5.5 By using the maximum area per sprinkler head allowed, this creates a cushion that will allow for flexibility in the case of future construction efforts, i.e., wall moves, high storage tank, etc.

8.1.5.6 Where field conditions necessitate any material change from approved plans, the change shall be submitted for approval.

8.1.6 Foam Quantity

8.1.6.1 The quantities of foam concentrates to be used shall be based on not less than 0.16 gpm/ft² (6.5mm/m²) and not less than 10 minutes, unless otherwise approved or stated by AHJ or Local Code.

8.1.7 Sprinkler Water Containment

8.1.7.1 Sprinkler Water Containment shall be provided to avoid unintentional discharge or release of hazardous chemicals and fire sprinkler water. Containment is primarily required for solvent rooms containing flammable and combustible liquids.

8.1.7.2 The designer shall consider and account for containment of sprinkler water and the worst-case discharge of Hazardous Production materials, in the case of an activation of the Fire Protection system.

8.1.7.3 The containment volume shall account for a minimum of 20 minutes sprinkler discharge duration of the Demand flow.

8.1.7.4 For rooms with chemical tanks or drums, in addition to the volume required by 20 minutes (minimum) of Demand Area flow, the containment volume shall be a function of the tanks/drums.

a) All tanks/drums are metal. The volume of the largest single chemical tank within the room shall be added to the containment volume.

b) Tanks/drums are plastic. The total volume of ALL plastic containers shall be added to the containment volume.

8.2 Specialty – Clean Agent Fire Extinguishing System

8.2.1 Occupancy Types and Minimum Design Concentration

Clean agent systems usually supplement automatic sprinklers and are not a substitute for them unless an exception is made within the appropriate occupancy-specific data sheet. The performance requirements of any clean agent fire extinguishing system inside a building are dependent upon the occupancy type, which identifies the nature of the hazard within the space. An occupancy with a high hazard will require a higher concentration of clean agent than an occupancy of a lower hazard. See **Table 4** and **Table 5** for typical occupancy types and minimum design concentrations. For any occupancy type not covered, contact Risk Management for guidance.

8.2.1.1 When the electrical equipment is de-energized upon activation of the clean agent extinguishing system, provide the minimum design concentration for an ordinary combustibles (Class A) fire.

8.2.1.2 When the electrical equipment is not de-energized upon activation of the clean agent extinguishing system or has a time-delayed power disconnect, provide the minimum design concentration for an energized electrical fire (Class C) in accordance with **Table 5** for the applicable equipment voltage. **Unless otherwise required by the AHJ or Local Code.**

8.2.1.3 **Table 6** provides information on physiological effects of inert gases agents covered by NFPA 2001 standard.

Table 4: Hazards Class and Occupancy Types

Hazard Class	Hazard Description	Occupancy
Class A	Surface fire hazards	
Class B	Ignitable liquid fire hazards	
Class C	Energized electrical/electronic equipment hazards with limited ordinary combustibles present	Electrical Room Computer Room IT Room Telecommunication Room SCADA server room

Table 5: Recommended Minimum Design Concentrations (Based on NFPA 2001)

Agent	Recommended Minimum Design Concentration, % (Resultant O2 Level ^{2,3} , %)		
	Class A	Class B ¹	Class C
IG-541 (Inergen)	34.2 (13.86)	40.3 (12.57)	38.5 (12..95)
IG-100 (Nitrogen)	37.2 (13.22)	41.9 (12.23)	41.9 (12.23)

¹ Class B values are for hydrocarbon fuels similar to n-Heptane. Class B minimum design concentration will vary for other ignitable liquid fuels, based on the information provided in the manufacturer's design, installation, operation, and maintenance manual.

² Resultant O2 level are reference from NFPA 2001, Table A.1.5.1.3(a) and Table A.1.5.1.3(c). With inert gas agents, an oxygen concentration of no less than 10 percent (sea level equivalent) is required for normally occupied areas.

³ Micron Global EHS requirement on the O2 levels shall be minimum 12% for inert gas system.

Table 6: Physiological Effects of Inert Gas Agents (Based on NFPA 2001)

Agent	No Effect Level (NOAEL) ¹ , %	Low Effect Level (LOAEL) ¹ , %
IG-541 (Inergen)	43	52
IG-100 (Nitrogen)	43	52

¹ Based on physiological effects in humans in hypoxic atmospheres. These vales are the functional equivalents of NOAEL and LOAEL values and correspond to 12% minimum oxygen for the no effect level and 10% minimum oxygen for the low effect level.

8.2.2 System Flow Calculation

8.2.2.1 System flow calculations shall be performed using a calculation method listed or approved by the AHJ. The system design shall be within the manufacturer's listed limitations.

8.2.2.2 Flow calculations along with the working plans shall be submitted to the AHJ for approval. The version of the flow calculation program shall be identified on the computer calculation printout.

- 8.2.2.3 Where field conditions necessitate any material change from approved plans, the change shall be submitted for approval.
- 8.2.2.4 Designs involving pre-engineered systems shall not be required to be provided with flow calculations.
- 8.2.2.5 Valves and fittings shall be rated for equivalent length in terms of pipe or tubing sizes with which they will be used. The equivalent length of the container valve shall be listed and shall include siphon tube, valve, discharge head, and flexible connector.
- 8.2.2.6 Piping lengths and orientation of fittings and nozzles shall be in accordance with the manufacturer's listed limitations.

8.2.3 System Discharge Time

- 8.2.3.1 Design inert gas clean agent systems to achieve 95% of the minimum design concentration listed in Section 8.2.1 in accordance with **Table 7**.

Table 7: Inert Gas System Discharge Time

Inert Gas Discharge Time (to achieve 95% of the min. design concentration)			
Agent	Class A	Class B	Class C
IG-541 (Inergen)	≤ 120 seconds	≤ 60 seconds	≤ 120 seconds
IG-100 (Nitrogen)	≤ 120 seconds	≤ 60 seconds	≤ 120 seconds

8.2.4 Duration of Protection

- 8.2.4.1 Provide the agent minimum design concentration within the protected enclosure for at least 10 minutes, or longer where necessary to ensure effective emergency response by trained personnel. Note that for Class C electrical fires, if the electrical equipment is not de-energized before the clean agent concentration falls below the minimum design value shown in **Table 5**, reignition is expected.
- 8.2.4.2 Maintain 85 percent of the minimum design concentration at the highest level of protected contents within the enclosure in accordance with **8.2.3.1**.

9.0 System Installation

9.1 Specialty – Foam Sprinkler System

9.1.1 Piping Arrangement

9.1.1.1 Connection Methods and fittings — refer to **Section 7.1**

9.1.1.2 Piping take-offs shall always occur at or above the horizontal plane. Connections below the horizontal plane are prone to scale accumulation. During an event, this scale may flow to the open heads and will prevent proper sprinkler head performance.

9.1.1.3 Ideally, take-offs should come from the top of the pipe and include the use of return bends.

9.1.1.4 Where possible, horizontal mains should be sloped upwards as they leave the riser room. This will allow for more complete draining and filling of the system.

9.1.1.5 The combination of sloping, proper placement of High Point Vents, and the installation of N2 Inerting Stations will extend the life of the system by minimizing corrosion potential and greatly improve the Operation and Maintenance of the overall systems.

9.1.2 Hangers and Seismic Bracing

9.1.2.1 Unless otherwise allowed by AHJ, piping for Fire Protection systems shall be supported independently of any other systems or utilities.

9.1.2.2 Hangers and seismic bracing shall be designed and installed using an approved computer design program.

9.1.2.3 Seismic design parameters shall be based on FM Global Data Sheet 2-8 "Earthquake Protection for Specialty Room Fire systems". Properly identifying the seismic zone is critical for proper design.

9.2 Specialty – Clean Agent Fire Extinguishing System

9.2.1 Piping Arrangement

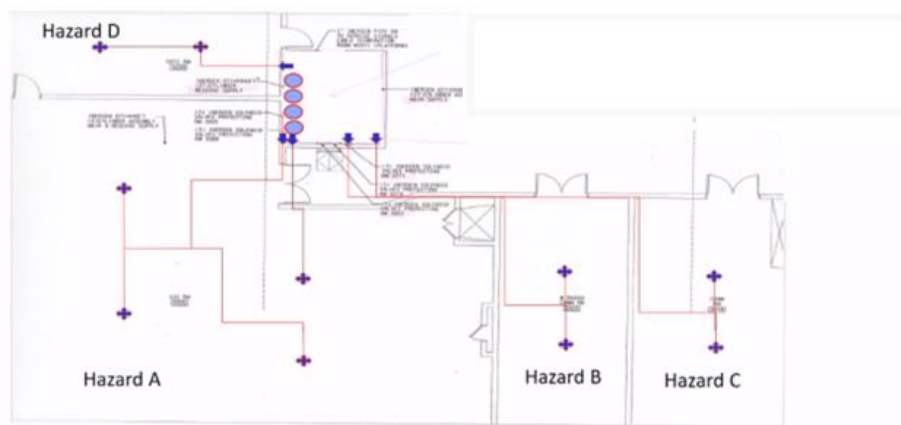
9.2.1.1 Connection Methods and fittings — refer to **Section 7.2**

9.2.1.2 Distribution of the clean agent fire extinguishing system shall be determined during actual design based on selected pre-engineered or engineered

system which comply with all NFPA, local code and AHJ requirements.

- 9.2.1.3 If multi-zone distribution is designed, the acceptable maximum number of zones shall not exceed 4 zones. Multi-zone system flow calculations shall be performed using a calculation method listed or approved by the AHJ. The multi-zone system design shall be within the manufacturer's listed limitations and submitted to the AHJ for approval. Refer to **Figure 9** for example of centralized multi-zone system.

Figure 9: Example centralized multi-zone system



9.2.2 Hangers and Seismic Bracing

- 9.2.2.1 Unless otherwise allowed by AHJ, piping for Fire Protection systems shall be supported independently of any other systems or utilities.
- 9.2.2.2 Hangers and seismic bracing shall be designed and installed using an approved computer design program.
- 9.2.2.3 Seismic design parameters shall be based on FM Global Data Sheet 2-8 "Earthquake Protection for Specialty Room Fire systems". Properly identifying the seismic zone is critical for proper design.

9.2.3 Detection and Activation

- 9.2.3.1 Provide automatic detection and activation for clean agent system using approved fire detection devices suitable for the protected hazard or occupancy as required to comply with all NFPA, local code and AHJ requirements.

- 9.2.3.2 Applicable more restrictive local code requirements at each location supersede any requirement contained within this section or applicable reference material (Section 12.2).
- 9.2.3.3 Detectors and others fire alarm component shall refer to "Global Facilities – Design & Construction Standard – I&C – Fire Alarm and HSSD System".
- 9.2.3.4 Clean Agent Fire Extinguishing System shall be designed as such any personnel within the enclosure shall be alert and evacuate immediately prior to actuation of clean agent. Refer to **Figure 10** for example and **Table 8** for recommended activation plan. Detail shall be determined during actual design.

Figure 10: Example Room Setup with Clean Agent Fire Extinguishing System

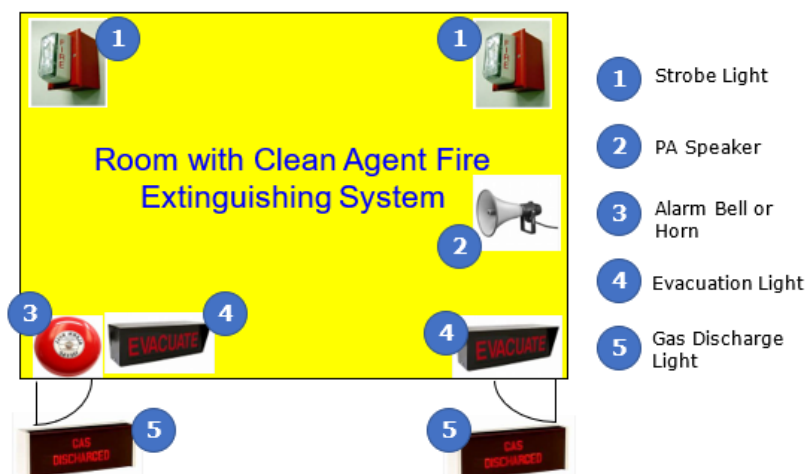


Table 8: Clean Agent Fire Extinguishing Recommended Activation Plan

Description	Detector 1 OR Detector 2 Trigger	Detector 1 AND Detector 2 Trigger
Dry Signal send to Local Panel	Yes	Yes
Local Panel Dry Signal to Main Fire Alarm Panel	Yes Supervisory Alert	Yes Alarm
Timer Delay	No	30 sec

Description	Detector 1 OR Detector 2 Trigger	Detector 1 AND Detector 2 Trigger
Visual & Audible Devices (Activation) 1. Bell or Warning Sound 2. Evacuation Announcement 3. Strobe Light Flash 4. Evacuate Light Flash 5. Gas Discharge Light Flash	1. Yes – Inside Room 2. Yes – Inside Room 3. Yes – Inside Room 4. Yes – Inside Room 5. No	1. Yes – Building Fire Alarm 2. Yes – Building Fire Alarm 3. Yes – Inside Room 4. Yes – Inside Room 5. Yes – Outside Room

9.2.4 Back-up Cylinder

9.2.4.1 It is recommended to provide minimum of one full replacement set on-site (warehouse within the same premises), at all times.

9.2.4.2 In addition to item 9.2.4.1, provide another set off-site and available within 12 hours (which can be consignment storage from supplier).

10.0 Quality Inspection and Testing

10.1 **Inspection and Test Plan (ITP)** – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

10.1.1 Piping

- Piping Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualification and Inspection
- Piping Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning and Flushing
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

10.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

10.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

10.2 **QA/QC Inspection & Punch List**– Prior to filling, leak testing, and pressure testing, QA/QC inspection & punchlist shall be performed on all installed piping and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards. Perform a final P&ID line check to ensure the installation has been properly reinstated after the pressure testing, cleaning, flushing, etc.

10.3 **Gross Leak Test** - Gross leak test shall be performed via compressed Nitrogen or Air at pressure up to 5 psig, on metallic piping only. Walk the entire system to verify that no leaks are present and make repairs to any leaks that are identified.

10.4 **Pneumatic Test** – Pneumatic test shall be performed for open-ended piping in a closed circuit for a period of 10 minutes at 40 psig. The pressure drop shall not exceed 20% of the test pressure at the end of 10 minutes.

10.5 Hydrostatic Pressure Test

10.5.1 **Foam Sprinkler System** - All piping shall be hydronic pressure tested at a minimum of 200 psig (13.8 Bar) pressure for a minimum of 2 hours, as required by NFPA 13 Section 25.1.

10.5.2 **Clean Agent Fire Extinguishing System** - Hydrostatically test any closed-section piping to a minimum of 1.5 times the maximum working pressure for 2 minutes if pneumatic test for clean agent fire extinguishing system is not possible.

Properly vent the system to remove air via all air vents, during the water filling process.

The actual hydronic pressure test procedure, including means, methods, testing pressures, and durations, shall be determined during the construction project based on project specific requirements. The procedure shall be reviewed and approved by Micron for use.

11.0 Startup Commissioning / Site Acceptance Test

11.1 Upon completion of the fire protection system installation, perform acceptance tests as required and outlined in NFPA 13, NFPA 16, NFPA 2001 and as required by the AHJ.

11.2 The equipment/components listed below shall be tested at a minimum, but not limited to the following:

11.2.1 Bladder Tanks

11.2.2 Check Valves / Backflow Preventers

11.2.3 Pressure Reducing Valves

11.2.4 Pressure Relief Valves

11.2.5 Foam Concentrate Proportioners

11.2.6 All Control Valves

11.2.7 Flow Switches and Supervisory Switches

11.2.8 Pre-Engineered or Engineered Clean Agent Fire Extinguishing System

12.0 Reference Specifications

12.1 NFPA

- NFPA 11 (2021), "Standard for Low-, Medium-, High- Expansion Foam"
- NFPA 13 (2019), "Standard for the Installation of Sprinkler Systems"
- NFPA 14 (2019), "Standard for the Installation of Standpipe and Hose Systems"
- NFPA 15 (2017), "Standard for Water Spray Fixed Systems for Fire Protection Systems"
- NFPA 16 (2019), "Installation of Foam-Water Sprinkler and Foam-Water Spray Systems"
- NFPA 25 (2020), "Specialty Room Fire Protection Systems"
- NFPA 30 (2021), "Flammable and Combustible Liquids Code"
- NFPA 318 (2018), "Standard for the Protection of Semiconductor Fabrication Facilities"
- NFPA 2001 (2018), "Clean Agent Fire Extinguishing Systems."

12.2 FM Global

The following are references made from FM Global Database

- DS 2-0, "Installation Guidelines for Automatic Sprinklers" ([LINK](#))
- DS 4-9, "Halocarbon and Inert Gas (Clean Agent) Fire Extinguishing System" ([LINK](#))
- DS 4-12, "Foam-Water Sprinkler Systems" ([LINK](#))
- DS 7-7, "Semiconductors Fabrication Facilities" ([LINK](#))
- 5130, "Approval Standard for Foam Extinguishing Systems"

13.0 Document Control

13.1 Approval: GLOBAL_FAC_GFTT_DOC_APPR

13.2 Notification

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

13.3 Final Viewer Access

All Micron sites Facilities Team Members

13.4 Retention

Adherence to the requirements specified in the
[**Global Facilities - Document Control Procedures**](#)

13.5 Review

Biennially (two years)

14.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	11/15/21
1	All Tables	All Table numbering has been revised due to added new table.	5/15/23
	Table 1	Change from NOVEC 1230 and FM200 recommended clean agent to IG-541 and IG-100.	
	Section 6	<u>6.2.1</u> Change from NOVEC 1230 and FM200 recommended clean agent to IG-541 and IG-100. Pre-action system maybe considered if IG-541 and IG-100 not accetable by AHJ and its use is subject to review and approval by Global Facilities <u>6.2.8</u> Added 6.2.8 and Table 2.	
	Section 7	<u>7.2.1</u> Added section 7.2.1 Storage Container <u>7.2.3.2</u> Remove the last bullet point. <u>7.2.3.3</u> Added 7.2.3.3 "For inert gas system incorporating a pressure reducing valve, use a pressure for piping and fitting wall thickness calculation upstream of the valve as indicated in Section 7.2.1.3. Downstream of the pressure reducing valve, use the maxixmum expected pressure bassed on the pressure reducing valve's settings" <u>7.2.3.4</u> Added 7.2.3.4 "Provide automatically operated pressure relief devices where valve arrangements result in closed piping sections. Design the pressure relief devices	

Revision #	Section Changed	Details of Changes	Date
		to operate at a pressure in accordance with the manufacturer's design, installation, operation, and maintenance manual and any applicable national or international codes and standards."	
	Section 8	<u>8.2.1.2</u> Added "unless otherwise required by the AHJ or local code" <u>8.2.1.3</u> Change 8.2.1.3 to "Table 5 provides information on physiological effects of inert gases agents covered by NFPA 2001 standard" <u>8.2.1.4</u> Removed 8.2.1.4 <u>Table 5</u> Change the table title to "Minimum Design Concentrations (Based on NFPA 2001)" Change from NOVEC 1230 and FM200 to IG-541 and IG-100 <u>Table 6</u> Added Table 6, "Physiological Effects of Inert Gas Agents (Based on NFPA 2002)" <u>8.2.3</u> Added section 8.2.3 System Discharge Time <u>8.2.4.2</u> Rectified typo "8.3.1" to "8.2.3.1"	
	Section 9	<u>9.2.4</u> Added 9.2.4 Back-up Cylinder	